

教學大綱 Syllabus

科目名稱：地圖學

Course Name:
Cartography

學年學期：115-1
Fall Semester, 2026

科目代碼：207056001
Course No.207056001



修別：
必

Type of
Credit:
Required

3.0

學分數
Credit(s)

50

預收人
數

Number
of
Students

課程資料

Course Details

開課單位：地二土測

Course Department: Land Economics/B/2

授課老師：范噶色

Instructor: STEPHAN VAN GASSELT

先修科目：無

Prerequisite(N/A)

上課時間：四234

Session: thu09-12

課程簡介

Course Description

Cartography is the technique, science, and art of transforming data into maps. These maps help us see and understand complex spatial patterns and relationships in a visual way.

Maps are everywhere --- they are part of our daily lives, often used without us even noticing. While traditional paper maps are still important, they have largely been replaced or supplemented by digital maps. However, the core idea of turning data into useful visual information remains the same. Thanks to the digital revolution, cartography has undergone major changes and successfully adapted to modern tools and technologies.

This course introduces students to **Digital Cartography** by focusing on how maps are made --- from choosing and organizing data, to filtering and interpreting it, to designing clear, effective maps. Students will explore these steps while creating their own maps using powerful digital tools.

Because of its ability to store, manage, analyze and visualize spatial data, **Geographic Information Systems (GIS)** plays a central role in this process. Throughout the course, students will use GIS to support their cartographic work and bring their map ideas to life.

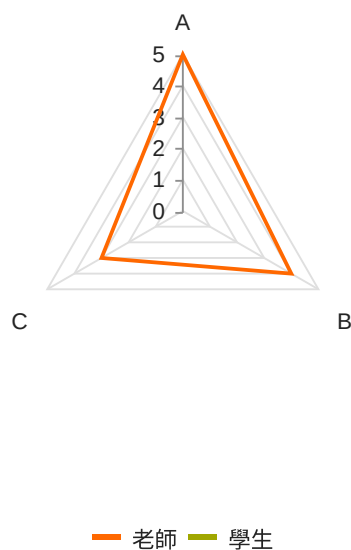
In addition to traditional cartographic concepts, the course introduces students to modern technologies, including **Machine Learning (ML)**, that are beginning to influence how maps are

made, interpreted, and automated. Students will gain foundational insights into how ML techniques can be used to classify spatial data, detect patterns, or assist in map generalization and feature extraction, helping them understand the growing role of intelligent systems in the future of cartography.

核心能力分析圖

Core Competence Analysis Chart

雷達圖



能力項目說明

- A. 具備地政相關專業知能
- B. 具備地政相關實務應用與創新之能力
- C. 具備領導與團隊合作並進之精神與國際觀

課程目標與學習成效

Course Objectives & Learning

Outcomes

This course provides a comprehensive introduction to the theoretical and practical foundations of cartography and map design in the digital age. It emphasizes how spatial data is transformed into effective visual communication through maps. Students will gain hands-on experience using Geographic Information Systems (GIS) and be introduced to state-of-the-art technologies, such as Machine Learning tools for mapping, that are reshaping the way spatial data is analyzed and visualized.



Map creation will focus on a variety of use cases, including physical maps, charts and plans, complex thematic maps incorporating relevant topics such as hazard mapping and Sustainable Development Goals (in particular SDG 11 and 15), and topological maps (network maps)

By the end of this course, students will be able to:

- Understand and apply core cartographic concepts, including scale, projection, generalization, and symbolization.
- Identify, evaluate, and select appropriate spatial data for specific mapping purposes.
- Design effective map concepts that reflect the intended message and audience.
- Transform raw data into higher-level, interpretable information through filtering, classification, and visual hierarchy.
- Produce digital maps using standard cartographic techniques and GIS tools that adhere to principles of effective visual communication.
- Use international cartographic terminology to describe and critique their map products.
- Gain foundational awareness of how AI and machine learning are used in cartography, e.g., in automating feature recognition, supporting data generalization, or enhancing spatial pattern detection.
- Critically assess the strengths and limitations of AI-supported mapping workflows.

課程屬性

Course Characteristics

級分說明 Score Description:

1關聯性最弱，2關聯性稍弱，3關聯性中等，4關聯性稍強，5關聯性最強。(1-5非百分比，而是Rank)
1 indicates weakest relevance, 2 indicates slightly weaker relevance, 3 indicates moderate relevance, 4 indicates slightly stronger relevance, and 5 indicates strongest relevance. (1-5 does not represent a certain percentage but indicates the rank).



課程屬性項目 Course Characteristics Items	級分 (1~5) Ranking	說明 Description
人文關懷課程 Humanity and Care Course	1	課程教學內容涉及學生人文關懷素養與態度之培養。 The course contents include the development of students' abilities and appreciation for humanistic care.
跨領域課程 Interdisciplinary Course	2	課程教學包含不同學門、領域之內容。 The courses include contents from different disciplines and fields.
社會參與課程 Social Engagement Course	1	課程教學內容涉及社會參與、服務等。 The course contents include social engagement and services.
總整課程 Capstone Course	4	係統整與深化學生歷年所學，並運用於實務操作與討論，藉以檢視和評估學生的整體學習成效。 The courses are designed to integrate and intensify students' academic results from previous years and apply them in practice and discussions for the purpose of reviewing and evaluating the overall effectiveness of students' learning.
永續課程 SDGs	3	課程主軸明確探討永續性之統整概念及/或聯合國永續發展目標及/或「生態」與「社會/文化/經濟」系統之間的相互依賴性。關聯度級分請點選連結說明。 A unit of instruction that has a primary and explicit focus on the integrated concept of sustainability, and/or UN Sustainable Development Goals, and/or the interdependence of ecological and social/cultural/economic systems. Please refer to the link of relevance score.
資訊判讀課程 Information Literacy Course	5	課程內容旨在教授媒體、網路相關專業資訊之真偽辨別與判讀能力。 The course would provide skills to evaluate the quality of information found on the media and the Internet.
AI 素養課程 AI Competence Course	3	將 AI 知能運用融入課程，引導師生與 AI 共學、協作，多元培養學生 AI 素養。 Integrate the application of AI knowledge into the curriculum, lead teachers and students to learn and collaborate with AI, and cultivate students' AI literacy.

每周課程進度與作業要求

Course Schedule & Requirements

Classes are 3 hours and take place in the GIS lab (270610). Due to the highly practical nature of this course, we will follow a hybrid approach with mixed theory and practice.



This course and all handout/upload material are provided in English, therefore a basic command of the English language will be required. Also, a basic understanding of spatial data and a feeling for spatial information and relationships are of advantage. For the technical aspects of this course, a basic knowledge of trigonometry and trigonometric identities, and a basic understanding of coordinate systems and transformations will be beneficial.

WEEK 1 (09/10). Introduction to Digital Cartography and Basics of GIS

CONTENTS. This week introduces the scope and purpose of digital cartography and distinguishes it from traditional mapmaking. You will become familiar with what a map represents in digital systems, the concept of spatial data, and the role of GIS software. The objective is to provide a conceptual foundation while setting up the digital environment for mapping.

READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. You attend an introductory lecture on the nature and roles of digital maps, followed by a hands-on lab exercise within a GIS environment with basic layer operations and an initial map design.

HOMEWORK. Readings and review.

WEEK 2 (09/17). Coordinate Systems, Projections and Map Scale

CONTENTS. This session focuses on geographic and projected coordinate systems, projection types, and their effects on spatial accuracy. You will learn how to choose and apply map projections and understand the importance of scale in spatial representation and interpretation.

READINGS. Course script and selected sections in Robinson et al. (1995), Snyder (1987), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. The lecture introduces map projections, scale concepts, and distortion principles. In the lab, you will reproject layers, observe differences between map projections and reference systems, and explore scale impacts on geography.

HOMEWORK: Start of Map Project 1: Population Density Thematic Map

WEEK 3 (09/24). Symbolization, Colour, and Visual Hierarchy

CONTENTS. This week covers visual variables such as colour, shape, size, and texture, along with symbol classification methods (e.g., quantile, natural breaks). Emphasis is placed on creating visual hierarchy and meaningful symbolization for thematic mapping.



READINGS. Course script and selected sections in Robinson et al. (1995), Bertin (2011), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. After a theoretical overview of symbol design and classification techniques, you will practice applying symbology in GIS, experimenting with different classification schemes and colour palettes to visualise population data.

HOMEWORK. Continue Map Project 1

WEEK 4 (10/01). Labels, Legends, and Map Composition

CONTENTS. Students explore cartographic composition, focusing on text labelling, map elements (legend, north arrow, scale), and visual balance. The goal is to produce clear, readable layouts that communicate spatial information effectively.

READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. The lecture discusses typefaces, label placement rules, and the function of cartographic elements. The lab is devoted to refining the thematic map layout, adding labels for cities, and finalising composition.

HOMEWORK. Finalise Map Project 1

WEEK 5 (10/08). Terrain Representation and Land Cover Data

CONTENTS. This week introduces elevation models and terrain visualisation techniques, including hillshading and contouring for physiographic and topographic maps. Students will also begin working with land cover raster datasets to understand physiographic mapping.

READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. The lecture covers elevation representation, contour logic, and land cover classification. The lab includes generating contours from Digital Elevation Models, styling hillshading, and importing land cover data.

HOMEWORK. Start of Map Project 2: Physiographic Map

WEEK 6 (10/15). Generalisation and Feature Extraction with Machine Learning

CONTENTS. This session explores generalisation techniques such as simplification and smoothing, and introduces basic machine learning for land cover classification. The aim is to automate the extraction of urban, water, and vegetation features.



READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. Students will experience vector generalisation methods, then use supervised classification to extract land cover types for a defined area. These features are converted into clean vector layers for further map use.

HOMEWORK. Continue Map Project 2

WEEK 7 (10/22). Map Refinement and Styling for Terrain Visualisation

CONTENTS. This week focuses on blending multiple terrain visualisation techniques and achieving expressive cartographic styles. Students will consolidate topographic features with elevation cues and meaningful symbology.

READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. Students finalise their physiographic map by integrating shaded relief, generalisation results, and proper labelling. Emphasis is placed on stylistic coherence and effective visual communication.

HOMEWORK. Finalise Map Project 2

WEEK 8 (10/29). Midterm Exam Week

WEEK 9 (11/05). Topology and Network Structures in Cartography

CONTENTS. Students are introduced to topological relationships and their use in transportation networks. The focus is on nodes, edges, and how spatial topology differs from visual or geographic fidelity.

READINGS. Course script and other sources.

ACTIVITIES. The lecture explains network logic, topological rules, and abstraction in metro maps. In the lab, students digitise station points and rail lines, ensuring correct topological structure.

HOMEWORK. Start of Map Project 3: Metro Network Topological Map



WEEK 10 (11/12). Spatial Adjustment and Network Geometry Control

CONTENTS. This session develops practical skills in aligning, snapping, and editing geometries while maintaining topological relationships. Students learn to optimise layout for readability without violating topology.

READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. After a lecture on geometric control, students modify and adjust metro networks, snapping nodes and refining layout while preserving logical connections between stations.

HOMEWORK. Continue Map Project 3.

WEEK 11 (11/19). Finishing the Network Map

CONTENTS. This week emphasises map finalisation and symbol simplification for abstract networks. The goal is to create a clean metro map with readable labels, coherent linework, and a functional layout.

READINGS. Course script and other sources.

ACTIVITIES. Students finish the metro map using abstraction techniques inspired by transit design standards. They apply symbology, finalise label placement, and prepare the map for submission.

HOMEWORK. Finalise Map Project 3

WEEK 12 (11/26). Digitizing and Working with Scanned Maps

CONTENTS. Students learn to georeference scanned historical maps and digitise vector features. Focus is on parcels and alignment to control points while maintaining topological consistency.

READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. The lecture covers georeferencing and digitising best practices. In the lab, students begin working with a historic parcel map, aligning it to base data and starting parcel digitisation.

HOMEWORK. Start of Map Project 4: Parcel Plan

WEEK 13 (12/03). Survey Constraints and Parcel Editing

CONTENTS. This session involves applying geometric constraints to parcel boundaries and editing nodes and edges while preserving clean topology. Students will learn to encode survey information and manage parcel attributes.



READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. Students complete parcel digitisation, manage shared edges, and adjust geometries to match survey-derived measurements. The lab includes layout preparation and styling for cadastral clarity.

HOMEWORK. Finalise Map Project 4.

WEEK 14 (12/10). Capstone Project Launch: Complex Thematic Map

CONTENTS. The capstone project introduces complex thematic mapping with multivariate symbology and multiple thematic layers combined with topographic basemaps. Students will work with earthquake point data, scale and colour symbols, and derive patterns from hotspot surfaces.

READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. After visiting multivariate and derived mapping, students begin compiling earthquake data from online catalogs. They explore magnitude-depth visualisation and generate kernel density maps to identify seismic clusters.

HOMEWORK. Start of Capstone Map Project: Complex-Thematic Map

WEEK 15 (12/17). Advanced Layouts and Capstone Integration

CONTENTS. This session supports the final integration of thematic layers, tectonic context, and cartographic refinement. Emphasis is placed on clarity, storytelling, and thematic coherence.

READINGS. Course script and selected sections in Robinson et al. (1995), Field (2018), Tyner (2010), and other sources.

ACTIVITIES. Students finalise their capstone maps by adding tectonic plate boundaries, labels, legends, and compositional elements. The focus is on exporting a polished product for submission.

HOMEWORK. Continuation of Capstone Map Project: Complex-Thematic Map

WEEK 16 (12/24). Final Exam Week

Finalization of Capstone Map Project: Complex-Thematic Map



授課方式

Teaching Approach

30%

講述
Lecture

20%

討論
Discussion

40%

小組
活動
Group
activity

10%

數位
學習
E-
learning

0%

其
他：
Others:

評量工具與策略、 評分標準成效

Evaluation Criteria

This course is developed around map projects. Additional homework assignments and bonus exercises will help to consolidate the obtained knowledge.

- The midterm examination will be a portfolio of online questions and/or a project covering the theoretical foundations of cartography.
- The final exam will be a mapping project. In order to complete the final project successfully, students will set up a mapping project, produce a digital map and describe their mapping concept and realization.

指定/參考書目

Textbook & References

All relevant material will be distributed during class. Modern cartography builds on traditional experience and knowledge in communication but applies modern tools and visualization techniques. For a good understanding, both aspects are needed: underlying concepts and modern tools.

- Bertin J (2010): Semiology of Graphics: Diagrams, Networks, Maps.- 456 pp., ESRI Press.
- Field K (2018): Cartography.- 576 pp., ESRI Press.
- Robinson, AH et al. (1995): Elements of Cartography.- 6th ed., 688 pp., Wiley.
ISBN 978-0471555797 (A good, though slightly out-dated introductory text).
- Snyder JP (1987): Map Projections -- A Working Manual.- United States Geological Survey Professional Paper 1395. U.S. Government Printing Office (Washington DC).
- Tyner, JA (2010): Principles of Map Design.- xii+259 pp., The Guilford Press (New York).
ISBN 978-1-60623-544-7.

已申請之圖書館指定參考書目



本課程可否使用 生成式AI工具

Course Policies on the Use of
Generative AI Tools

有條件開放使用:It is encouraged to use generative AI tools to develop ideas for cartographic concepts and to identify essential datasets and visualization styles. Generative AI must not be used for in-class assignments and tests/exams.
Conditional Permitted to Use

課程相關連結

Course Related Links

The International Cartographic Association (<https://icaci.org>) and its commissions on all sorts of cartographic topics (<https://icaci.org/commissions/>).



課程附件

Course Attachments

課程進行中，使用智慧型手機、平板等隨身設備

To Use Smart Devices During the
Class

是
Yes

授課教師Office Hours及地點

Office Hours & Office Location

My office is located on the 15th floor, room number 271535 and I follow an open-door policy. Students may drop by in case of questions, remarks or any other matter of discussion.



The teaching assistant for Cartography will be announced in due time.

教學助理基本資料

Teaching Assistant Information

